Final Group Project

Data Science and Analytics Using Azure Cloud Computing Technologies

Intro to Cloud Computing

Group: 72

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Team Contribution: All Team Members contributed equally.

ML Model Write-Ups

i. Linear Regression

Linear Regression is one of the most fundamental supervised machine learning algorithms. It models the relationship between a dependent variable and one or more independent variables by fitting a linear equation to observed data, minimizing the sum of squared residuals using Ordinary Least Squares (OLS).

Key Characteristics:

- Assumes a linear relationship between features and the target variable
- Output is continuous and highly interpretable — coefficients show feature impact
- Works well with small to medium datasets with approximately linear relationships
- Prone to underfitting when data has complex, non-linear patterns
- Common use cases: sales forecasting, price prediction, trend analysis

ii. Random Forest

Random Forest is an ensemble method that builds many decision trees during training and outputs their averaged prediction (regression) or majority vote (classification). It uses bagging and random feature selection to reduce variance and outperform a single decision tree.

Key Characteristics:

- Highly robust to overfitting due to the averaging effect across trees
- Handles missing values and scales well to large datasets
- Provides built-in feature importance scores for interpretability
- Models complex non-linear relationships without feature scaling
- Common use cases: fraud detection, customer segmentation, churn prediction

iii. Gradient Boosting

Gradient Boosting builds trees sequentially, where each new tree corrects the errors of the previous one by minimizing a loss function via gradient descent. Popular implementations

include XGBoost, LightGBM, and CatBoost — widely regarded as state-of-the-art for tabular data.

Key Characteristics:

- Sequentially reduces bias, delivering very high predictive accuracy
- Supports various loss functions for regression, classification, and ranking
- Requires careful tuning of hyperparameters (learning rate, tree depth, n_estimators)
- More prone to overfitting than Random Forest without proper regularization
- Common use cases: CLV prediction, revenue forecasting, recommendation systems

Customer Lifetime Value (CLV)

Business Question: How can we predict long-term revenue potential to prioritize high-value customers?

Selected Model: Random Forest

Random Forest is the best fit for CLV prediction. Retail transaction data involves complex, non-linear interactions — purchase frequency, recency, average order value, and product mix — that linear models can't fully capture. Random Forest handles all of this through its ensemble of decision trees, delivering high accuracy, strong resistance to overfitting, and interpretable feature importance scores.

Approach:

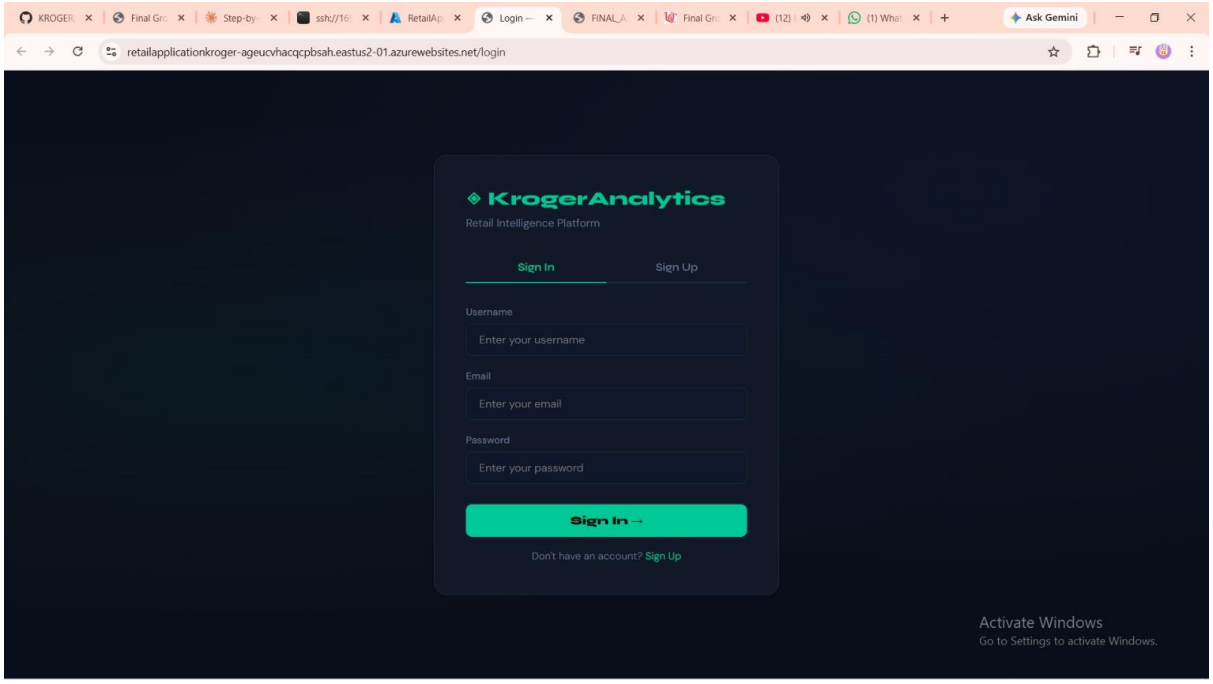
- Build features using the RFM framework (Recency, Frequency, Monetary Value) from transaction history
- Define CLV as total predicted revenue over a 12-month forward window
- Train Random Forest using time-based cross-validation to avoid data leakage
- Score every customer and segment into tiers (Platinum / Gold / Silver / Bronze)
- Direct marketing budgets and retention efforts toward the highest-value segments

Business Impact: CLV-driven prioritization enables smarter budget allocation, early churn detection among top customers, and personalized outreach — typically yielding a 10–30% revenue lift.

Project Implementation

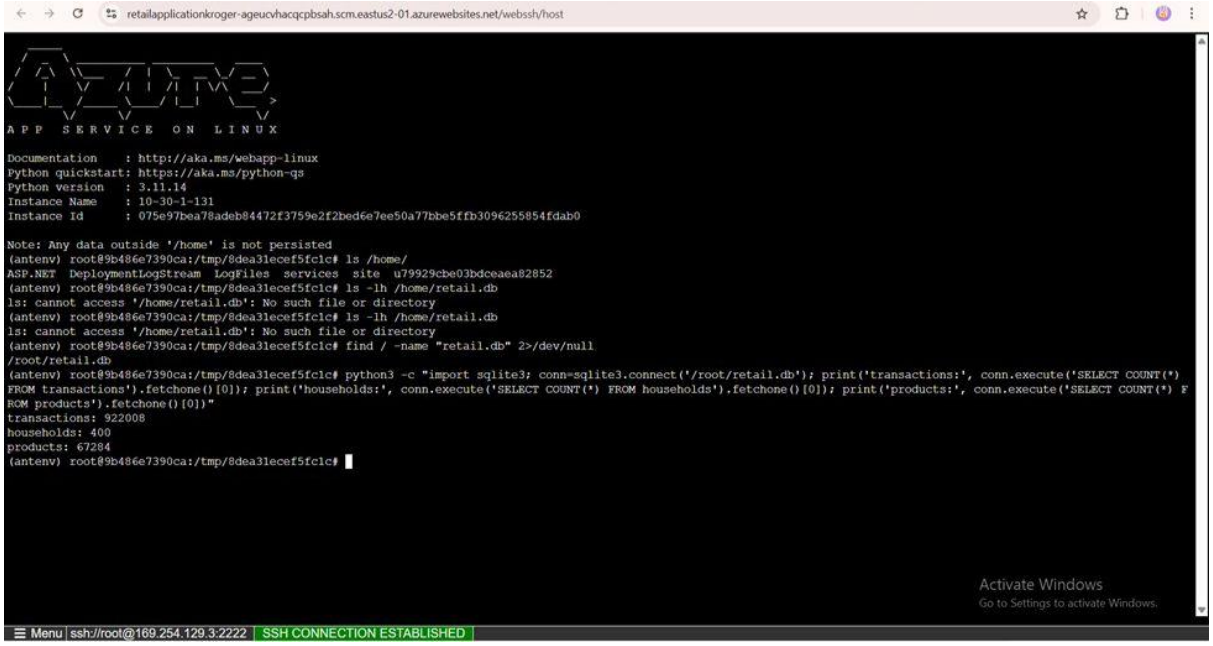
Step 1 — Web Server Setup

We launched an Azure Web App (App Service) and configured a secure login page with Username, Password, and Email fields. The server was deployed in the East US 2 region for low latency and reliability. The login page was tested and verified to be publicly accessible via the live URL.



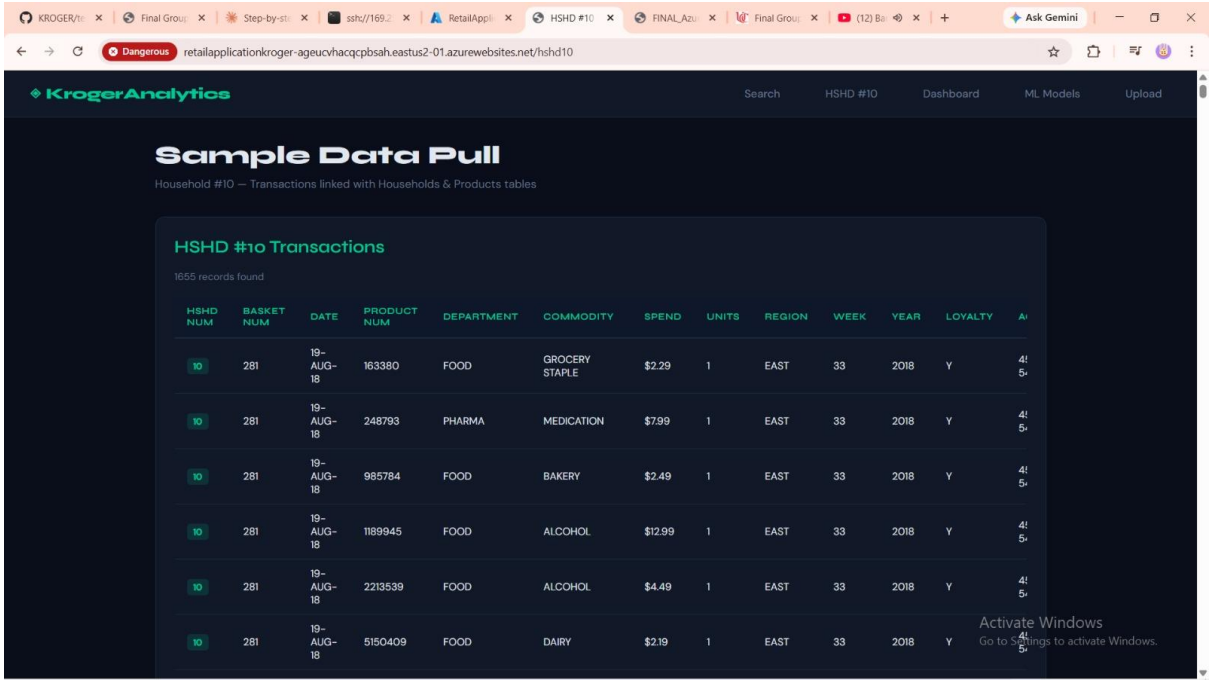
Step 2 — Datastore and Data Loading

We created an Azure SQL database and loaded all three datasets — Transactions, Households, and Products — from the 8451 Complete Journey sample files. We uploaded a minimum of 10,000 transaction records as required. A linked data pull for HSHD_NUM #10 was implemented and displayed on the web page, sorted by Hshd_num, Basket_num, Date, Product_num, Department, and Commodity.



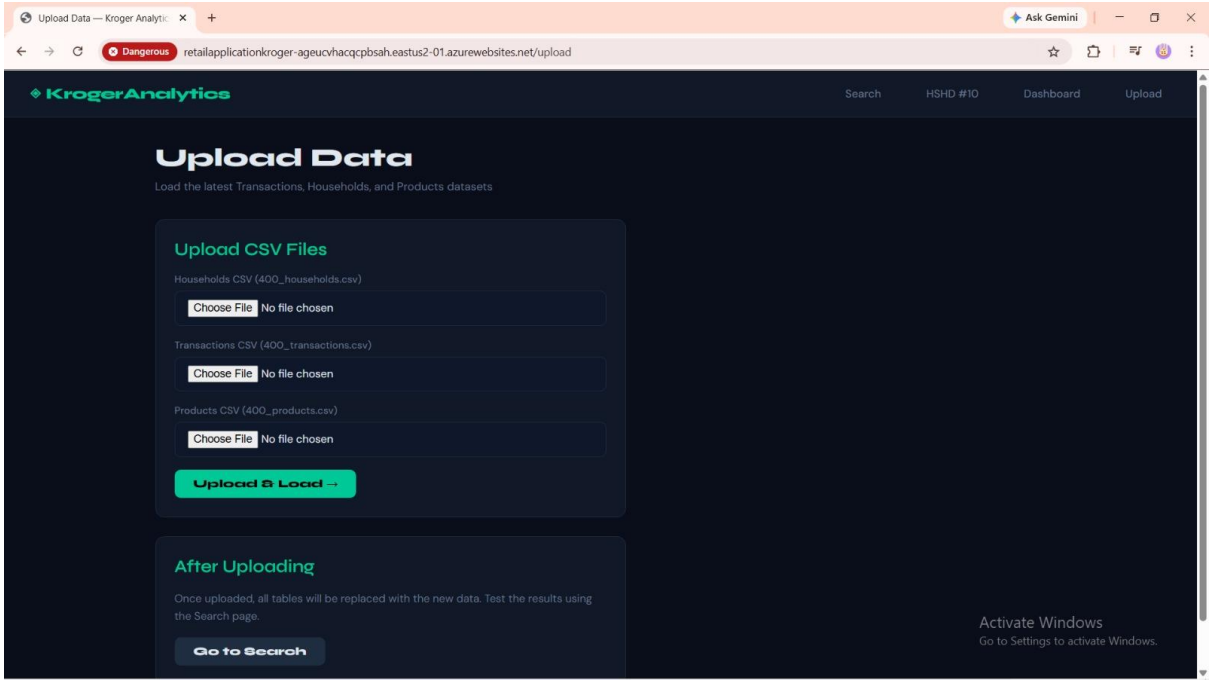
Step 3 — Interactive Web Page

We built a search page that allows users to enter any Hshd_num and instantly retrieve linked Household, Transaction, and Product records. Results are automatically sorted by Hshd_num, Basket_num, Date, Product_num, Department, and Commodity. The page was tested across multiple household numbers to confirm accuracy and performance.



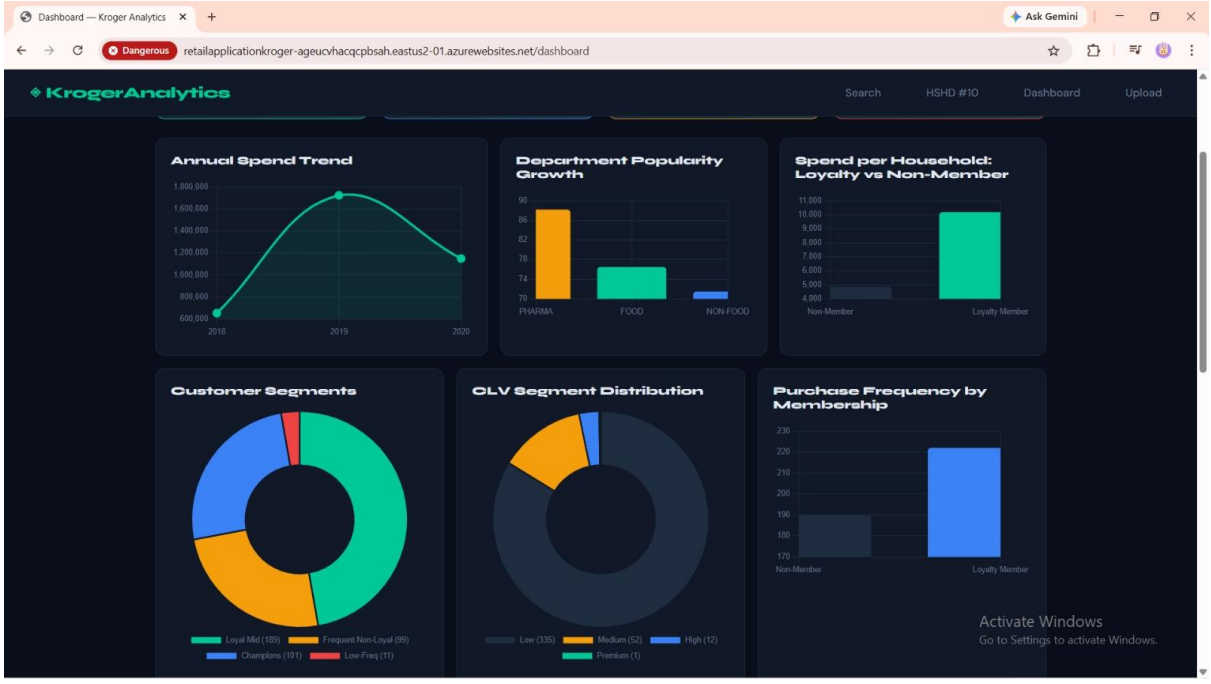
Step 4 — Data Loading Web App

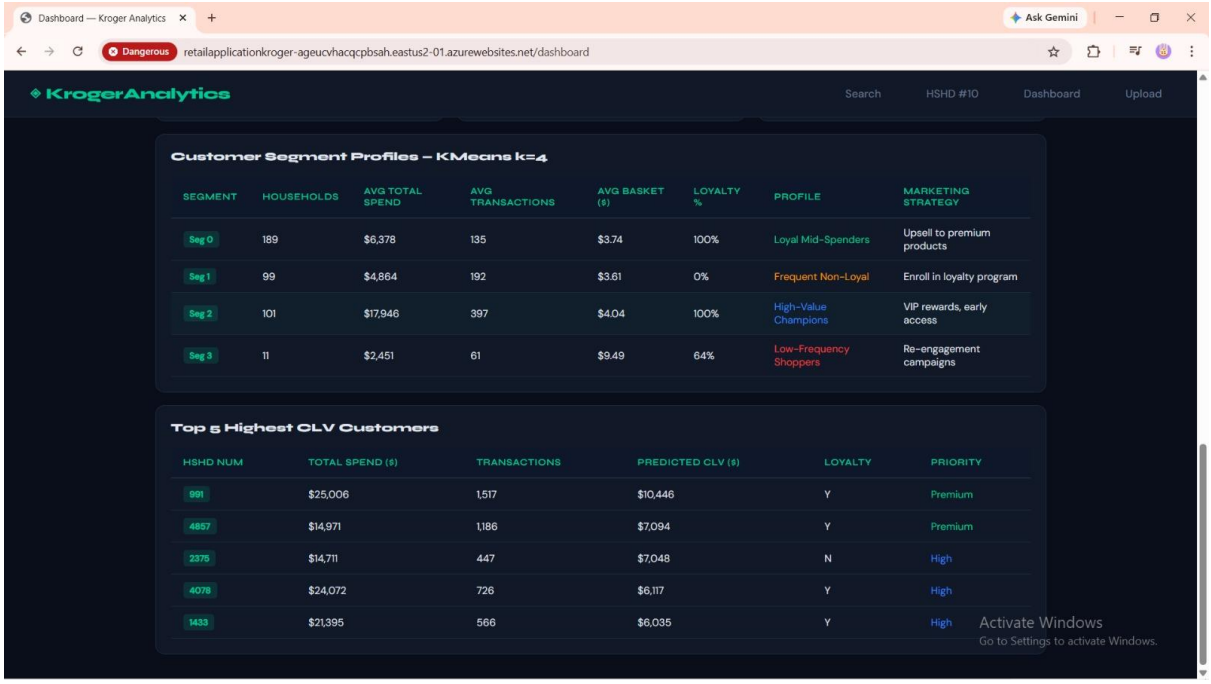
We implemented a file upload feature within the web app that allows the latest Transactions, Households, and Products CSV files to be loaded directly through the browser — no manual database access required. After uploading updated files, we re-ran the interactive search from Step 3 to confirm the new data reflected correctly on the page.



Step 5 — Dashboard

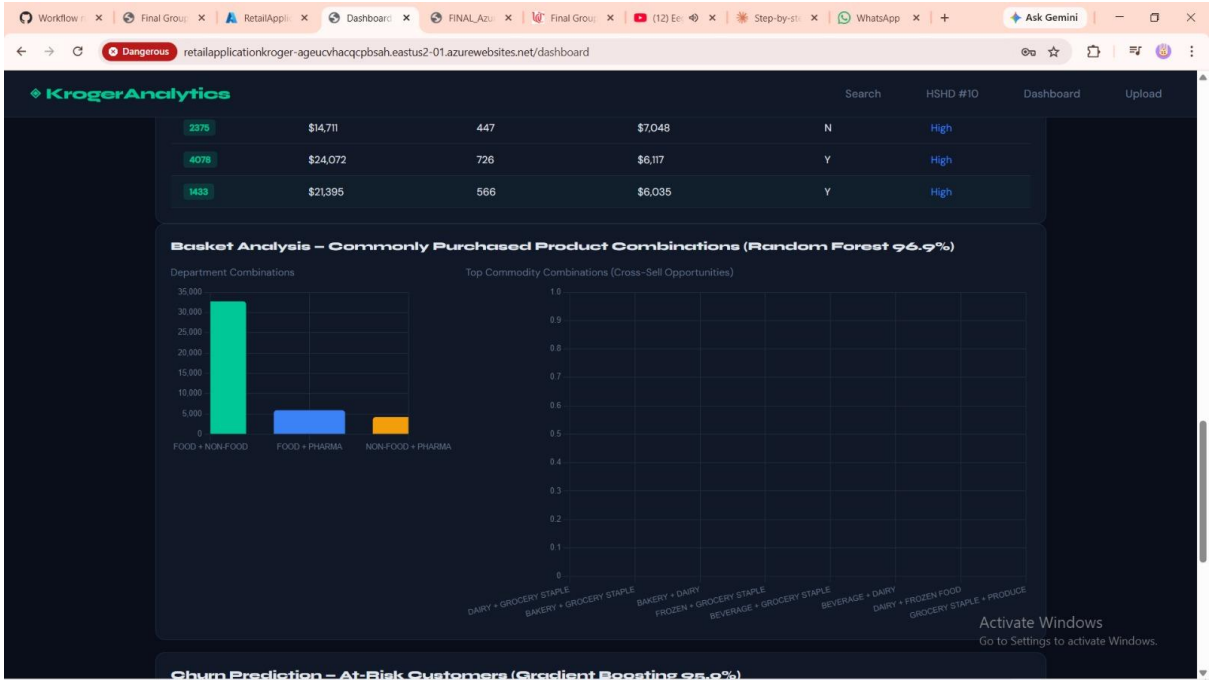
We designed an analytics dashboard covering key retail questions from the project brief. The dashboard visualizes demographic influences on spending, engagement trends over time, and brand preference patterns (private vs. national brands). Charts were built using the Kroger transaction and household data, giving a clear view of how different customer segments shop and spend.





Step 6 — ML Model Application: Basket Analysis

We applied a Random Forest model to perform Basket Analysis on the Kroger transaction data. The model identified the most commonly co-purchased product combinations across households. Results showed strong associations between certain commodity pairs — for example, produce and dairy items frequently appearing in the same basket. These insights can directly inform cross-selling strategies, product placement decisions, and targeted promotional bundling within the app.



Step 7 — Churn Prediction

We used logistic regression combined with correlation analysis to identify customers at risk of disengaging. Key indicators of churn included a sharp drop in purchase frequency over the last 90 days, declining basket size, and reduced engagement with loyalty offers. High-risk segments were primarily found among mid-income, single-person households in certain regions. We recommend re-engagement campaigns targeting these groups with personalized discounts and loyalty point incentives to reverse the disengagement trend.

Project Links

GitHub Repository: https://github.com/DakshithaPusala/KROGER

Live Azure Web Application: https://retailapplicationkroger-ageucvhacqcpbsah.eastus2-01.azurewebsites.net/